Shyama Sayi Gummadi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.96
Chettinad Vidhyashram, RA Puram, Chennai	Class XII (CBSE)	2018	85.00%
Hindu Senior Secondary School, Indira Nagar, Chennai	Class X (CBSE)	2016	9.60%

SCHOLASTIC ACHIEVEMENTS

- Qualified IIT JAM 2025 in Mathematics
- Qualified GATE 2025 in Civil Engineering
- Cleared the ISI B.Math entrance examination, 2018
- Completed a Minor in Computer Applications alongside a B.Tech in Civil Engineering at NIT Tiruchirappalli
- Secured grade S (the highest grade) in Data Structures and Applications at NIT Tiruchirappalli

RELEVANT COURSES

- B.Tech: Data Structures and Applications, Basics of Programming, Mathematical Foundations of Computer Science, Operating Systems, Data Analytics, Information Security, Numerical Techniques, Probability, Statistics and Linear Programming
- M.Tech: Applied Computational Lab, Advanced Statistical Methods in Hydrology, Remote Sensing and GIS Applications in Civil Engineering

PROJECTS

Satellite Data Pipeline for Reservoir Monitoring (SWOT) | *Remote Sensing & GIS, IIT Hyderabad*

- Built a Python pipeline that retrieves satellite time-series data from the NASA Hydrocron REST API for 57 candidate river reaches across 3 Godavari Basin reservoirs
- Handled failed and empty API responses, ranked reaches with a custom scoring metric, and generated comparative analytics with pandas and Matplotlib
- Filtered reaches geospatially in Google Earth Engine using buffer analysis on the SWORD river database

Markov Chain Time-Series Modelling of River Discharge and Sediment | *Advanced Statistical Methods, IIT Hyderabad*

- Implemented two published Markov chain models from scratch in Python on multi-decade daily river records from the Central Water Commission
- Built quantile-based state discretisation, transition probability matrices and stationary distributions, and optimised model weights with linear programming (SciPy)
- Identified strong first-order dependence in daily series and multi-year memory in annual runoff

Integrated Flood Modelling and Inundation Mapping | *Hydraulic & Hydrologic Simulation Lab, IIT Hyderabad*

- Built a 27-subbasin HEC-HMS rainfall-runoff model and HEC-RAS 1D/2D hydraulic models for 10 design-storm and historical flood scenarios
- Extracted cell-level results from HDF5 model outputs, and validated the models through volume-conservation checks (error below 0.2%) and domain-boundary checks

SKILLS

- Languages: Python, R, C++
- Libraries: pandas, NumPy, SciPy, Matplotlib, Seaborn
- Tools: Google Colab, RStudio, Google Earth Engine, HEC-HMS, HEC-RAS
- Concepts: data structures, REST API data retrieval, time-series analysis, statistical modelling, linear programming

POSITIONS OF RESPONSIBILITY

- Teaching Assistant, IIT Hyderabad (Jul 2025-Present): TA for 4 courses, Fluid Mechanics, Hydraulic Engineering and their respective labs
- Evaluated assignments and carried out exam invigilation duties for the theory courses
- Taught and demonstrated lab experiments to 50+ students, including minor losses in pipe fittings and water hammer in pipes
- Class Representative, B.Tech Civil Engineering, NIT Tiruchirappalli (Jul-Dec 2021): represented a class of 100+ students as the point of contact with faculty for academic schedules, announcements and student concerns